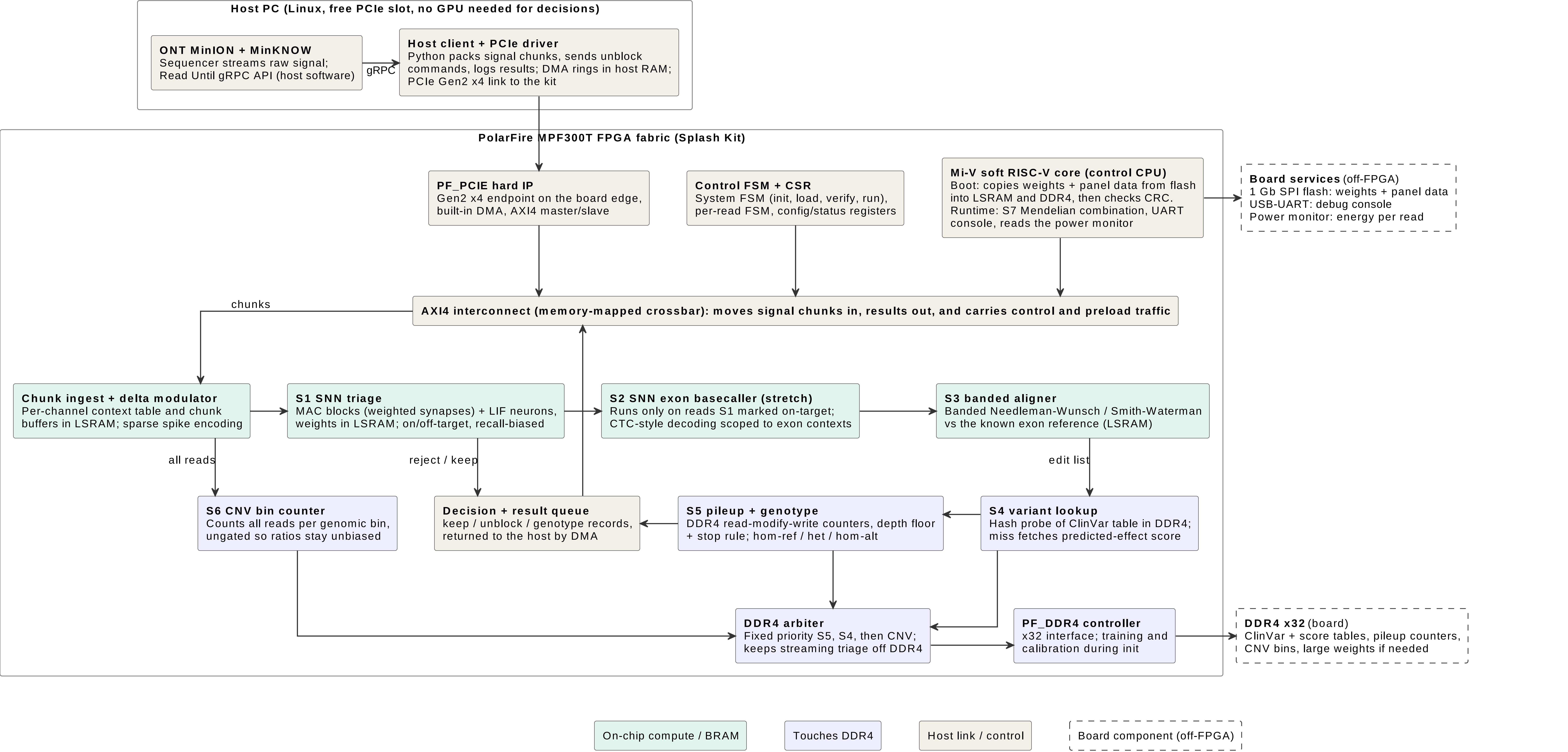


System block diagram: SNN-gated real-time exon variant detection on the PolarFire Splash Kit (MPF300T)



How it works

- MinKNOW streams raw signal to the host client, which DMAs chunks over PCIe into the FPGA.
- Ingest buffers each chunk per channel and delta-modulates it into spikes.
- S1 classifies the read. Off-target reads are rejected at once and an unblock goes back to the host.
- On-target reads continue to S2 (stretch goal), then S3 aligns them to the exon reference.
- S4 looks up each edit in DDR4. S5 updates depth counters and stops when depth is sufficient.
- S6 counts all reads per genomic bin in parallel, before any triage gating.
- The decision queue returns keep / unblock / genotype records over PCIe.
S7 on the Mi-V core combines parental genotypes.

Memory plan

On-chip LSRAM / uSRAM (about 20.6 Mbit in total):
SNN weights, per-channel neuron state, chunk buffers, exon reference.
DDR4 x32 (large, slower): ClinVar and score tables, pileup counters, CNV bins, and stage 2 weights if they do not fit on-chip.
SPI flash: boot copy of weights and panel data, loaded at init.
Host link: PCIe DMA carries chunks in and results out.